Survival Analysis of Concrete Highway Bridge Decks in Oregon Utilizing LASSO and Stepwise-Variable Selection

Jael Wettach-Glosser[1]; Avinash Unnikrishnan, Ph.D.[2]; and Thomas Schumacher, Ph.D., P.E., M.ASCE[3]

Abstract: This research studied the relationship between Oregon national bridge inventory (NBI) data as well as additional design/detailing variables and concrete highway bridge deck performance. Cox proportional hazards regression using least absolute shrinkage and selection operator (LASSO) and stepwise-variable selection were employed to determine the contribution of independent variables gathered from the NBI database. The stepwise approach produced a model with fewer variables compared with when LASSO was used. The hazard ratios (HR) of those variables retained in the stepwise approach were consistent with HR obtained from LASSO. The HR were also found to be intuitive. For example, cast-in-place bridge decks on steel bridges without a wearing surface or deck protection have an increased risk of deterioration while bridge decks in dry climate zones far from the ocean on prestressed concrete bridges have a decreased risk of deterioration. A separate refined dataset of 400 bridge decks was assembled by manually going through bridge drawings to gather information on relevant design/detailing variables. Bridge decks were carefully selected using a stratified sampling procedure based on design period and climate zone. Kaplan-Meier survival curves were constructed to study the impact of these design/detailing variables on concrete highway bridge deck performance. For example, bridge decks with concrete cover less than 1 in. (25 mm) have a lower survival probability than bridge decks with more cover and bridge decks that are slender have a higher survival probability than other bridge decks. **DOI: 10.1061/(ASCE)BE.1943-5592.0001606**.

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Introduction and Background

The presented research studied the parameters that influence concrete bridge deck deterioration, and might ultimately be used to improve performance monitoring and develop optimal preservation schedules for concrete highway bridge decks.

In order to improve the estimation of bridge performance, researchers have developed various methods that utilize different models for predicting bridge deterioration. The most widely used models for predicting bridge deterioration are stochastic models. Of stochastic models, the most common ones utilize Markov chains and are implemented in bridge management systems such as Pontis and BRIDGIT (Agrawal et al. 2010). In 2003, Morcous et al. (2003) attempted to improve the use of Markovian deterioration models by efficiently categorizing bridge deck environments. The researchers utilized genetic algorithms to determine environmental categories through the best combination of bridge deck deterioration parameters such as highway class, region, average daily traffic (ADT), and percentage of truck traffic. The approach was applied to bridge deck data gathered from the Ministére des Transports du Québec and used to create new Markovian deterioration models. Setunge and Hasan (2011) employed a Markovian model to bridge data gathered from the Roads Corporation of Victoria in Australia. The study found that bridge decks deteriorate faster than other bridge components.

While practical and easy to implement, Markovian models do have some limitations. For example, they do not consider condition history when assessing deterioration (Madanat et al. 1997). In addition, estimating transition probabilities when considering maintenance and repair actions is not trivial (Madanat et al. 1995). Because of the shortcomings associated with Markovian models, other approaches for predicting the deterioration of bridges have been explored. For example, an ordered probit model was developed by Madanat et al. (1995) to link deterioration to select independent variables. To do this, the ordered probit model was applied to bridge deck data gathered from the Indiana State Bridge Inventory database, which was composed of approximately 5,700 bridges. The independent variables used in this study were wearing surface, structure type, climate region, bridge age, ADT, highway classification, deck width, span length, number of spans, and skewness (Madanat et al. 1995). The results of the model found that the proposed method was more realistic than the state-of-the-art Markovian transition models. Similarly, two years later Madanat et al. (1997) used the same data to test for heterogeneity and the Markovian assumption that deterioration is independent of condition history. Using a probit model for bridge deck deterioration, the study found that results improve when heterogeneity and state dependence are considered. To complement these findings, Bulusu and Sinha (1997) developed a Bayesian model and a binary probit model for the estimation of condition states considering heterogeneity and state dependence for bridge substructure data gathered from the State of Indiana.

[1]Structural Engineer I, DOWL, 1111 Main St., Suite 401, Vancouver, WA 98660. ORCID: https://orcid.org/0000-0001-5953-323X. Email: jwettachglosser@dowl.com

[2]Associate Professor, Civil and Environmental Engineering, Portland State Univ., 1930 SW 4th Ave, Suite 200, Portland, OR 97201. Email: uavinash@pdx.edu

[3]Associate Professor, Civil and Environmental Engineering, Portland State Univ., 1930 SW 4th Ave, Suite 200, Portland, OR 97201 (corresponding author). ORCID: httsp://orcid.org/0000-0003-0118-9119. Email: thomas.schumacher@pdx.edu

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In an effort to assert the validity of using Markov chain models for predicting bridge condition, Morcous (2006) evaluated the impact of the two Markov chain assumptions on predicting bridge deck performance. The first assumption was that "bridge inspections are performed at predetermined and fixed time intervals, i.e., a constant inspection period," the second assumption was that "future bridge condition depends only on the present condition and not on the past condition i.e., state independence" (Morcous 2006). Using statistical tests on Markov chain models created from bridge deck data gathered from the Ministère des Transports du Québec, the study found that variation in the inspection period might result in a 22% error in service-life prediction. In addition, the study also found that the state independence assumption is acceptable with a 95% level of confidence. All the methods mentioned previously represent state-based stochastic models that predict the probability that the condition state of a bridge component will change at a given time depending on a list of independent variables.

Alternatively, time-based models can be used to predict the time it takes a bridge component's condition state to change. An important benefit of time-based models is that they can consider both censored and uncensored data (Greene 1997). Censored data correspond to the case in which the value of an observation is only partially known. In the context of bridge deck condition data, censoring can be a result of missing or unobservable condition ratings in an inspection report. Uncensored data by contrast are completely observable. The section "Time-in-Condition Rating" describes the types of censorship found in national bridge inventory (NBI) bridge deck condition ratings in detail. On that note, Mauch and Madanat (2001) developed stochastic duration models using the semiparametric Cox proportional hazards technique for bridge decks in the Indiana Bridge Inventory database. The independent variables used to predict condition time-in-state in the study were type of wearing surface, deck type, climate region, bridge age, ADT, highway classification, deck width, number of spans, and number of lanes. To expand on that study, Mishalani and Madanat (2002) developed a probabilistic time-based duration model that indicates the significant influence of age, traffic loading, environmental factors, structure type, highway class, and wearing surface type on bridge deck deterioration. The study also developed a methodology to accurately determine transition probabilities of state-based models from corresponding time-based models, which allows for better deterioration predictions. Sobanjo et al. (2010) studied the deterioration patterns of NBI Florida bridge components in various condition states to develop a reliability model. Using this model, survival and hazard functions were developed that show bridge categories deteriorate faster depending on age and roadway type. Similarly, Tabatabai et al. (2011) created a reliability model for NBI bridge decks in the State of Wisconsin. Using the Akaike information criteria, a hypertabastic failure time model was selected over Weibull, log logistic, and lognormal models. The results found that deck area, type of structure, and ADT are all important factors influencing bridge deck performance. Five years later, Tabatabai et al. (2016) expanded the hypertabastic failure model to perform a survival analysis on NBI bridge decks in Minnesota, Wisconsin, Michigan, Ohio, Pennsylvania, and New York. The analysis included the NBI parameters type of structure, deck surface area, age, and ADT. The results of the analysis indicated that ADT and deck surface area significantly affect reliability and failure rates for bridge decks in all six states.

In this paper, the authors present two separate analyses, which they applied to two different datasets. Similar to Mauch and Madanat (2001), Cox proportional hazards models were used to study the impact of NBI variables on concrete highway bridge deck performance. In order to expand on the previous work presented in the literature, the models created in this paper included a larger set of independent variables determined from the NBI to get a better understanding of the parameters that influence concrete highway bridge deck performance. In order to create robust models from these variables, two variable selection approaches, least absolute shrinkage and selection operator (LASSO) and stepwise, were employed. The authors also created a new refined dataset of 400 bridge decks focusing on design/detailing variables. The dataset was assembled by carefully selecting the bridge decks using a stratified sampling procedure and manually going through bridge drawings. Kaplan-Meier survival curves were created to study the impact of these design/detailing variables on concrete bridge deck performance. The inclusion of design/detailing variables, which has not been previously done, sheds new light on the driving factors of concrete highway bridge deck deterioration. Both the Cox proportional hazards model and Kaplan-Meier survival curves account for censorship in the used datasets.

Data and Data Sources

Two Oregon-specific datasets were created. The first dataset, referred to as "NBI Dataset," is composed of 3,319 concrete highway bridge decks and included information on variables gathered and derived from the Oregon NBI. The second dataset, referred to as "Refined Dataset," consists of 400 bridge decks distributed amongst nine climate zones and three design periods, and contains information on design/detailing variables collected from the Oregon Department of Transportations' (ODOT) bridge data system (BDS). The design periods used in this study are periods of time in which different design standards were used. On recommendation by ODOT engineers, the periods "before 1950," "1950–1970," and "after 1970" were selected because ODOT has found a significant difference in observed concrete bridge deck performance. The independent variables that were included in these two datasets were selected based on discussions with ODOT personnel and evidence from the literature. To quantify deterioration in bridge decks, time-in-condition-rating (TICR) was used as the performance metric and dependent variable. TICR is defined as the number of years a bridge deck is assigned the same NBI condition rating (CR) before it is assigned a lower CR. Since the effects of the independent variables can differ depending on a bridge deck's CR, only TICR with a CR value of 7 were used in the analyses presented in this paper. A CR value of 7 was chosen because it is the most common value and thus has the most available TICR data. Details regarding how TICR is computed and some associated data preprocessing are discussed in the subsequent section.

Time-in-Condition Rating

TICR was added to both datasets as the dependent variable for each bridge deck. TICR is calculated from NBI CR for the years of available inspection records (1992 to 2016) and gives a measure of how condition changes throughout a bridge deck's life (Nasrollahi and Washer 2015). Since a bridge experiences multiple CR throughout its observed life, multiple TICR occurrences can exist. Fig. 1 shows a CR record for a fictitious bridge deck, illustrating several examples of TICR occurrences. It can be observed that due to the nature of the data, censorship occurs, which is when the value of an occurrence is only partially observable (Leung et al. 1997). For the case of TICR, censorship occurs for three reasons: (1) CR before 1992 and after 2016 are unknown, (2) there are missing CR records, and (3) CR increases from one year to the next, which is assumed to be due to maintenance action, herein defined broadly as "action that increases the condition rating of a bridge deck" (Ghonima et al.

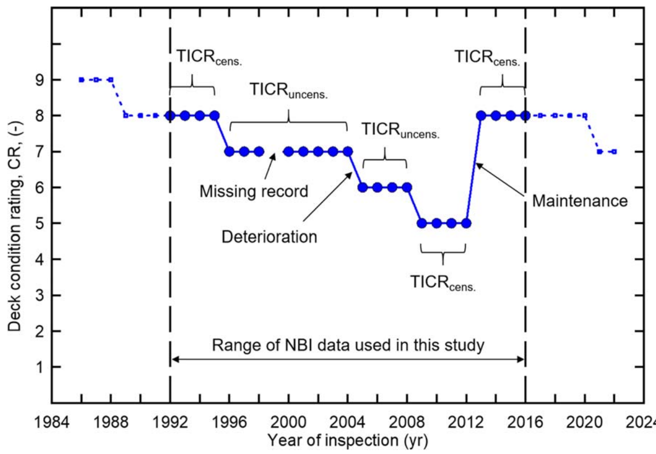

Fig. 1. CR record of fictitious bridge deck for available NBI data (1992–2016).

2018). For all of these cases, a minimum value for TICR can be computed but it is inappropriate to assume that this TICR is accurate because it is only observable partially. This type of censorship is referred to as right-censored. Because of the uncertainties associated with censored TICR, any statistics attempting to describe TICR without considering censoring significantly underestimates the actual values. This is an important consideration for NBI-based data since typically more than 70% of the TICR are right-censored. The following TICR can be computed from Fig. 1:

- TICR = 4, CR = 8, censored since CR is unknown before 1992,
- TICR = 9, CR = 7, uncensored since fully observable and CR decreases (the missing CR record would be interpolated following the procedure described subsequently),
- TICR = 4, CR = 6, uncensored since fully observable and CR decreases,
- TICR = 4, CR = 5, censored since CR increases due to maintenance action,
- TICR = 4, CR = 8, censored since CR is unknown after 2016.

In order to compute TICR for this study, some preprocessing was performed following Ghonima et al. (2018). For bridge decks with three years or less of missing records between recorded CR, missing data was filled in by a customized interpolation procedure. This procedure followed the subsequent rules:

- For the case where missing data existed between two CR with the same values, the missing records were assigned the same as the adjacent available CR values. For example, the missing record for 1999 in Fig. 1 would be replaced by a CR value of 7 data points.
- In the case where missing CR records were between two different CR values, two rules applied. First, for three years of missing CR, the first and third CR were assigned the adjacent available CR values. The second (or center) CR value was assigned randomly to either of the adjacent available CR values. Second, if two subsequent CR values were missing, the assignment was simply split between the adjacent available CR.

If more than three subsequent CR records were missing, no action was taken, which meant the TICR occurrences on either side of the missing records were considered censored [for more details regarding this procedure, see Ghonima et al. (2018)].

Calculating descriptive statistics for TICR is complicated due to the presence of censoring in the data. Means to treat censorship properly are discussed in the following sections.

Table 1. Summary statistics for continuous independent variables in the NBI Dataset

Variable name	Minimum	Mean	Maximum
Deck Area: computed from NBI Items 49 and 51	344 ft^2 (32 m^2)	8,839 ft^2 (821 m^2)	10,102,567 ft^2 (938,560 m^2)
log(ADTT/Lanes on Structure)	−1.82	2.03	3.79

NBI Dataset

The NBI is a comprehensive source of information for bridges with a span of at least 20 ft (6.20 m) in the United States that includes 116 data items that describe the characteristics and condition over time of each structure (FHWA 2018). For this study, a dataset was created of NBI concrete highway bridge decks in Oregon containing variables considered influential to bridge deck deterioration. To select only decks made of concrete, only concrete cast-in-place and precast deck panels were selected from the NBI using the variable Deck Structure Type (NBI Item 107) as a filter. In total, there were 5,242 NBI bridges with concrete highway bridge decks in Oregon in 2016. The number of bridge decks included in this dataset was further reduced as a result of missing data in the independent and dependent variables. In addition, only TICR with a CR value of 7 were included in the model. The reason for this was to create a model based on the most common condition type and to minimize overall variance in the model, since different CR have been found to behave quite differently (Ghonima et al. 2018). Once the bridge decks with missing information were removed, the final NBI Dataset was compiled, comprising 3,319 bridge decks.

For the bridge decks included in the NBI Dataset, independent variables considered influential to bridge deck deterioration were selected and derived from the NBI. The variables included were selected based on discussions with ODOT personnel as well as insights from the literature. In total, nine variables were selected directly from the NBI while five variables were derived from NBI Items (see Tables 1 and 2), which are Deck Area, Distance from Seawater, log(ADTT/Lanes on Structure), Climate Zone, and Design Period. To determine Bridge Deck Area, Structure Length (NBI Item 49) was multiplied by Bridge Roadway Width (NBI Item 51). The average Bridge Deck Area was found to be 8,839 ft^2 (821 m^2). In this study, ADTT (NBI item 109) was divided by Lanes on Structure (NBI item 28) and then the log taken; the average of this ADTT-related variable is 2.03. Distance from Seawater was calculated from the global position coordinates of the bridge deck (NBI Items 16 and 17). Some 98% of bridges are more than 0.621 mi (1 km) away from the ocean. To determine the Climate Zone of each bridge deck, the global position of each bridge deck was merged with climate zones identified by Taylor and Hannan (1999) (Fig. 2). Most bridges are located in the Willamette Valley, which corresponds to Zone 2. The Design Period variable was defined based on groups recommended by ODOT personnel ("Before 1950," "1950–1970," and "After 1970") and using the more recent year recorded in the variables Year Built or Year Reconstructed (NBI Items 27 or 106, respectively). The majority of bridges were built between 1950 and 1970. The rest of the variables included in this study were gathered directly from the NBI. The following are some observations: nearly two-thirds of bridges in Oregon are maintained by the State Highway agency; the majority of bridges are located in rural areas; nearly 85% of bridges are either simple or continuous span concrete or prestressed concrete bridges; the majority of bridge decks were constructed cast-in-place; over half of the bridge decks have a bituminous wearing surface; nearly 90% of bridges decks do not have a membrane while over 85% of bridge decks do not have deck

Table 2. Summary statistics for categorical independent variables in the NBI Dataset

Variable name	Categories	Frequency	Percentage
Distance from seawater: computed from NBI Items 16 and 17	1 = <0.621 mi (1 km)	52	1.6
	2 = >0.621 mi (1 km)	3,267	98.4
Climate zone: determined from NBI Items 16 and 17 and Taylor and Hannan (1999)	1 = Coast	449	13.5
	2 = Willamette Valley	1,343	40.5
	3 = Southwestern Valleys	565	17.0
	4 = Northern Cascades	52	1.6
	5 = High Plateau	25	0.8
	6 = North Central	293	8.8
	7 = South Central	223	6.7
	8 = Northeast	255	7.7
	9 = Southeast	114	3.4
Design period: determined from NBI Items 27 or 106 and recommendation from ODOT	1 = Before 1950	339	10.2
	2 = 1950–1970	1,538	46.3
	3 = After 1970	1,442	43.4
Maintenance responsibility (NBI Item 21)	1 = State highway	2,112	63.6
	2 = County highway	1,088	32.8
	4 = City/municipal highway	86	2.6
	11 = State park/forest/reservation	21	0.6
	21 = Other state	1	<0.1
	25 = Other local	10	0.3
	26 = Private	1	<0.1
Functional classification of inventory route (NBI Item 26)	1 = Rural principal arterial-interstate	352	10.6
	2 = Rural principal arterial-other	556	16.8
	6 = Rural minor arterial	368	11.1
	7 = Rural major collector	667	20.1
	8 = Rural minor collector	267	8.0
	9 = Rural local	370	11.1
	11 = Urban principal arterial-interstate	254	7.7
	12 = Urban principal arterial-other freeways or expressways	58	1.7
	14 = Urban other principal arterial	180	5.4
	16 = Urban minor arterial	145	4.4
	17 = Urban collector	65	2.0
	19 = Urban local	37	1.1
Kind of material and/or design (NBI Item 43A)	0 = Other	1	<0.1
	1 = Concrete	534	16.1
	2 = Concrete continuous	984	30.0
	3 = Steel	399	12.0
	4 = Steel continuous	87	2.6
	5 = Prestressed concrete	1,114	33.6
	6 = Prestressed concrete continuous	154	4.6
	7 = Timber	46	1.4
Type of design and/or construction (NBI Item 43B)	1 = Slab	721	21.7
	2 = Stringer/multibeam/girder	1,652	49.8
	3 = Girder and floorbeam	48	1.4
	4 = Tee beam	253	7.6
	5 = Box beam/girders	309	9.3
	6 = Single/spread box beam/girders	34	1.0
	7 = Frame	32	1.0
	9 = Deck truss	25	0.8
	10 = Thru truss	50	1.5
	11 = Deck arch	37	1.1
	12 = Thru arch	5	0.2
	13 = Suspension	1	<0.1
	15 = Lift	4	0.1
	16 = Bascule	1	<0.1
	17 = Swing	1	<0.1
	19 = Culvert	1	<0.1
	22 = Channel beam	145	4.4
Inspection frequency (NBI Item 91)	6	6	0.2
	12	111	3.3
	23	1	<0.1
	24	3,201	96.4

Table 2. (*Continued*.)

Variable name	Categories	Frequency	Percentage
Deck structure type (NBI Item 107)	1 = Cast-in-place	2,679	80.7
	2 = Precast panels	640	19.3
Type of wearing surface (NBI Item 108A)	0 = None	82	2.5
	1 = Monolithic concrete	990	29.8
	2 = Integral concrete	31	0.9
	3 = Latex concrete	236	7.1
	5 = Epoxy overlay	140	4.2
	6 = Bituminous	1,804	54.4
	8 = Gravel	28	0.8
	9 = Other	8	0.2
Type of membrane (NBI Item 108B)	0 = None	2,943	88.7
	1 = Built-up	40	1.2
	2 = Preformed fabric	247	7.4
	3 = Epoxy	22	0.7
	8 = Unknown	62	1.9
	9 = Other	5	0.2
Deck protection (NBI Item 108C)	0 = None	2,831	85.3
	1 = Epoxy	446	13.4
	2 = Galvanized	2	0.1
	3 = Other coated reinforcing	8	0.2
	4 = Cathodic protection	4	0.1
	8 = Unknown	28	0.8

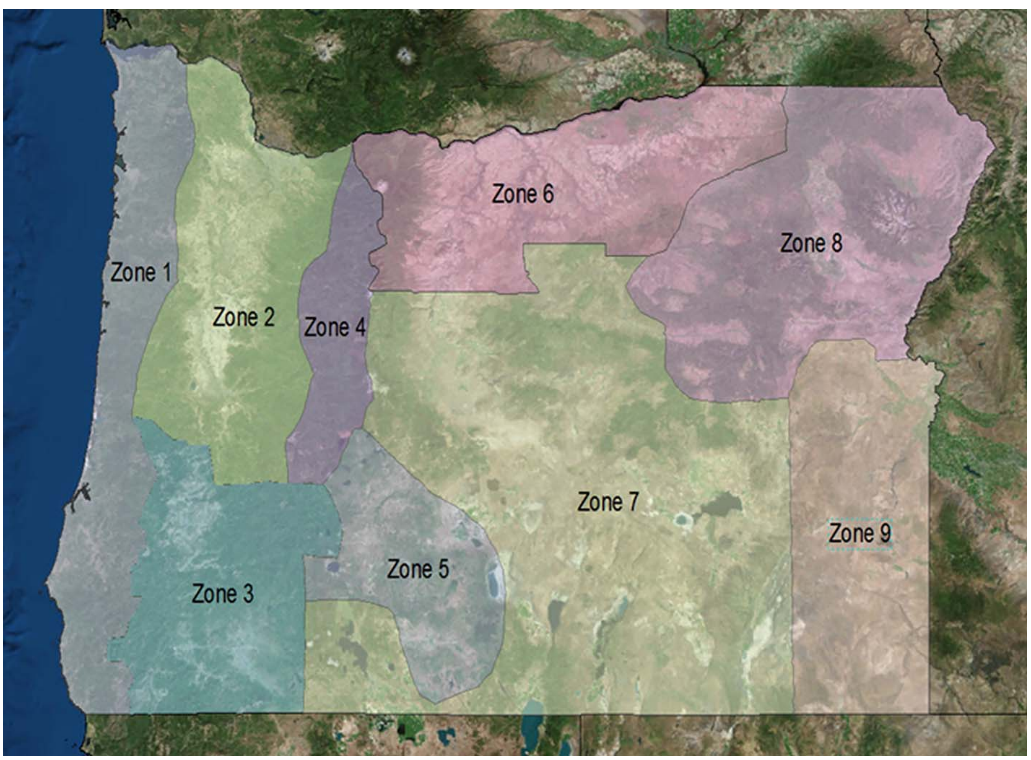

Fig. 2. Oregon climate zones according to Taylor and Hannan (1999). (Map data from Esri, DigitalGlobe, GeoEye, i-cubed, USDA FSA, USGS, AEX, Getmapping, Aerogrid, IGN, IGP, swisstopo, and the GIS User Community.)

protection. For consistency, names of the NBI variables and groups follow FHWA terminology (FHWA 1995).

Refined Dataset

The NBI Dataset only contains information on the general characteristics of bridges and their associated bridge decks. To get a more refined picture of bridge deck deterioration, examining design/detailing variables that describe bridge decks in further detail is important. Since design/detailing data are not available in the NBI database, a separate dataset referred to as "Refined Dataset" was created that includes design/detailing variables believed to influence concrete bridge deck performance. In order to create this dataset, bridge deck drawings gathered from ODOT's BDS were reviewed. The variables determined from these drawings were concrete cover, rebar type, rebar spacing, and deck slenderness.

Table 3. Number of bridge decks in each climate zone and design period group with complete design/detailing information

	Design period		
Climate zone	1	2	3
1	15	15	20
2	15	14	23
3	15	18	17
4	8	14	18
5	3	7	13
6	15	15	18
7	14	16	20
8	13	17	16
9	3	17	21

Table 4. Summary statistics for independent variables in the Refined Dataset

Categorical variables	Categories	Frequency	Percentage
Concrete cover	1 = Less than 1 in. (25 mm)	46.0	11.5
	2 = 1–2 in. (25–51 mm)	282	70.5
	3 = 2–3 in. (51–76 mm)	72.0	18.0
Transverse rebar spacing	1 = Less than or equal to 10 in. (305 mm)	112	34.0
	2 = Greater than 10 in. (305 mm)	217	66.0
Rebar type	1 = Black	336	84.2
	2 = Epoxy-coated	63.0	15.8
Deck slenderness	1 = Less than or equal to 0.05	80.0	20.2
	2 = 0.05–0.15	279	70.3
	3 = Greater than 0.15	38.0	9.60

The deck slenderness variable was determined by dividing the deck thickness by the smaller of the spaces between the beams or the space between bents. Although the BDS is an extensive database, not all bridge decks, particularly for old bridges, had drawings available. In addition, collecting data in this method was extremely time consuming. Therefore, instead of collecting design/detailing data for all concrete highway bridge decks in Oregon, a subset was established, consisting of randomly selected decks from groups based on the climate zone and design period (see Table 2 for details). The complete breakdown of selected bridge decks by climate zone and design period can be seen in Table 3. In total, four design/detailing variables from 400 bridge decks were selected from the BDS to create the Refined Dataset (Table 4). Some observations of this dataset are as follows: over 70% of bridge decks in the Refined Dataset have a concrete cover of 1–2 in. (25–51 mm). Two-thirds of bridge decks in the dataset have transverse rebar spacing greater than 10 in. (254 mm). Nearly 85% of bridge decks have black rebar, while the majority have a slenderness between 0.05 and 0.15.

Methodology

Regression Model for NBI Dataset

To identify the influence of the independent variables in the NBI Dataset on concrete highway bridge deck performance, a survival analysis was performed utilizing Cox proportional hazards regression. In Cox regression, the coefficients are estimated for each independent variable in a hazard function. This hazard function can be written as (Mauch and Madanat 2001)

$$h(t) = \lambda_0(t) \times e^{\beta_1 X_1 + \ldots + \beta_k X_k} \tag{1}$$

where t = the survival time; $h(t)$ = the hazard function determined by a set of k covariates ($X_1, X_2, \ldots, X_k$) and represents the expected number of events per unit of time; β_1 through β_k = the coefficients that measure the impact of the covariates; and $\lambda_0(t)$ = the baseline hazard function that corresponds to the hazard when all covariates are equal to zero. When the logarithm of this hazard function is taken, the Cox model can be written as a linear function of the variables X_i, from which the coefficients β_i can be determined (Sullivan 2016). By taking the antilog of the estimated regression coefficients, hazard ratios (HR) can be calculated, which describe the effect of each variable on survival. In this study, survival corresponds to the bridge deck CR remaining the same, given the current CR and set of independent variables. An HR > 1 indicates that there is an increased probability that CR decreases; in other words, there is an increased risk of deterioration. Analogously, an HR < 1 indicates that there is a decreased risk of deck deterioration.

Different approaches are available to select a subset of important variables to arrive at the final model. A common approach is to study the p-values of each independent variable and gradually remove variables with p-values higher than a cutoff value either one-by-one or as a group and perform statistical tests to carefully vet the decision to remove. Such an approach can be time consuming and is often unreliable in the presence of multicollinearity. LASSO is a variable selection technique based on the L1 regularization principle (Tibshirani 1996). In the Cox proportional hazard regression model, coefficients are estimated by minimizing the negative log likelihood function, $l(\beta)$. LASSO works by adding a penalty term that corresponds to the sum of the absolute value of coefficient magnitude, $\sum_{j=1}^{K} |\beta_j|$, as follows:

$$LL = -l(\beta) + \lambda \sum_{j=1}^{K} |\beta_j| \tag{2}$$

LASSO shrinks the coefficients of unimportant variables to zero and retains only the important variables. The λ term is a penalty factor. In this research, the authors use the glmnet package in *R* to implement LASSO (Tibshirani 1997; Friedman 2010). The glmnet package also performs N fold validation where data is randomly divided into N subsets or folds, and the model is trained on $N-1$ folds and tested on the remaining fold. This process is repeated N times, each time a different set is chosen as the testing fold. In this research, the results obtained from the LASSO-based variable selection procedure are compared with the results obtained from the standard backward stepwise-variable selection procedure. To perform stepwise-variable selection, the MASS package in *R* was used (Venables and Ripley 2002). In this procedure, the goal was to arrive at a subset of variables that minimizes the Akaike Information Criteria (AIC):

$$\text{AIC} = -2[l(\beta)] + 2K \tag{3}$$

where K = the number of covariates in the final model.

Survival Curves for Refined Dataset

Survival curves were created to visualize the impact of design/detailing variables on bridge deck survival probability over time for the Refined Dataset. To do this, the Kaplan-Meier estimator (Kaplan and Meier 1958) was used, which determines the probability that a bridge deck survives (or is assigned the same CR) past each time interval. In order to judge these survival curves, p-values were generated for the survival curves using the log-rank test (Sullivan 2016; Rich et al. 2010). A p-value ≤ 0.05 suggests that at least two survival curves are significantly different from each other at the 95% confidence level. Since the Refined Dataset has significantly less data then the NBI Dataset, a regression analysis could not be performed.

Results and Discussion

Regression Model for NBI Dataset

In order to develop the regression model and to consider all of the available data, the groups of the categorical independent variables were converted to dummy variables and a reference group was chosen. Table 5 shows the reference groups selected for each of the independent variables in the NBI Dataset.

The Cox proportional hazards model created using LASSO selected 52 out of the possible 88 independent variables to describe concrete highway bridge deck performance. To describe the effect of each categorical independent variable on bridge deck performance, HR were calculated for the important variables in the model (see Table 6). In addition, to validate the independent variables selected using the LASSO method, an additional model was created that utilized variable selection. To compare the results of the different models, HR were calculated and are presented in Table 6. Recall that HR < 1 and HR > 1 indicate decreased and increased probabilities of bridge deck deterioration, respectively. Unlike the LASSO results, the HR calculated using stepwise regression have p-values associated with them that indicate that the HR are significantly different from 1.

Discussion of Continuous Variables

Of the continuous variables, only log(ADTT/Lanes on Structure) was removed from the backward selection model. HR for this variable was found to be slightly greater than 1 for the LASSO model, which indicates that as it increases so does the risk of bridge deck deterioration. The conclusion that bridge decks with higher log(ADTT/Lanes on Structure) experience higher deterioration risk is consistent with Fleischhacker et al. (2020), Hatami and Morcous (2011), Agrawal et al. (2010), and Kim and Yoon (2010). In contrast, HR for the deck area variable are less than 1 for all models, which indicates that as bridge deck area increases the risk of deterioration decreases. However, the HR for this variable are only slightly less than 1, which suggests that the influence on bridge deck performance is not prominent.

Table 5. Reference dummy variables

Independent variable	Reference dummy variable
Climate zone	2 = Willamette Valley
Design period	3 = After 1970
Deck area	N/A
Distance from seawater	N/A
log(ADTT/lanes on structure)	N/A
Maintenance responsibility (Item 21)	1 = State highway
Functional classification of inventory route (Item 26)	1 = Principal arterial-interstate
Kind of material and/or design (Item 43A)	1 = Concrete
Type of design and/or construction (Item 43B)	2 = Stringer/multibeam/girder
Inspection frequency (Item 91)	24 months
Deck structure type (Item 107)	1 = Cast-in-place
Type of wearing surface (Item 108A)	6 = Bituminous
Type of membrane (Item 108B)	0 = None
Deck protection (Item 108C)	0 = None

Discussion of Categorical Variables

Distance from Seawater. The results from the LASSO model show that bridge decks less than 1 km from the ocean have HR $>$ 1, which indicates an increased risk of deterioration compared with bridges decks further from the ocean. This is consistent with Stewart and Rosowsky (1998) and Vu and Stewart (2000) who state that bridges within 0.621 mi (1 km) are exposed to chlorides, which increases bridge deck deterioration.

Climate Zone. Zones 1, 5, and 7 have HR < 1, which indicate a decreased risk of deterioration compared with Climate Zone 2. In contrast, Zone 3 has an HR > 1, which suggests an increased risk of deterioration compared to Zone 2. The higher risk of deterioration for bridge decks in Climate Zone 3 might be associated with the harsher climatic conditions of this region.

Design Period. Results indicate that for all models, bridge decks built before 1950 and bridge decks built between 1950 and 1970 have an increased risk of deterioration compared with ones built after 1970. This is most likely because of the improvements made in bridge design specifications and construction but age might also have an effect. It should be noted, however, that not all studies agree on whether increasing age increases or decreases deterioration. In fact, a recent study found that with increasing age of a concrete bridge deck it is less likely to exhibit high deterioration (Ghonima et al. 2020).

Maintenance Responsibility (NBI Item 21). Results from all of the models show that the only important agencies in comparison to state highway agency are county highway and city. Bridge decks maintained by both of these agencies have a decreased risk of deterioration compared with state-maintained bridge decks.

Functional Classification of Inventory Route (NBI Item 26). Bridge decks on rural and urban roads show a decreased risk of deterioration when compared with bridge decks on rural principal arterial/interstate routes for all models. These results make sense considering that all other routes have less traffic than rural principal arterial/interstates.

Kind of Material and/or Design (NBI Item 43A). Bridge decks in conjunction with prestressed concrete, prestressed concrete continuous, timber, or other bridge superstructure systems show a decreased risk of deterioration compared with concrete bridges. In contrast, bridge decks supported by a steel bridge superstructure have an increased risk of deterioration for all models. These HR are intuitive as they show that bridge decks associated with simple-span prestressed concrete bridge superstructures show a decreased risk of deterioration. The reason for this is the absence of negative bending moment regions, which create tension stresses in a bridge deck and thus increase the potential for cracking.

Type of Design and/or Construction (NBI Item 43B). Bridge decks associated with slab, girder and floorbeam, box beam/girder, single/spread box beam/girder, and lift bridges have a decreased risk of deterioration compared with stringer/multibeam/girder bridges in the LASSO model. In contrast, bridge decks supported by deck truss, thru arch, suspension, swing, and channel beam bridges have an increased risk of deterioration in the LASSO model. For the stepwise model, bridge decks on girder and floorbeam and box beam/girder bridges have a decreased risk of deterioration while bridge decks on channel beam bridges have an increased risk of deterioration.

Inspection Frequency (NBI Item 91). Results indicate that compared with bridge decks inspected every 24 months, which is the normal inspection interval, bridge decks inspected every 6 months have an increased risk of deterioration in the LASSO model. In

J. Bridge Eng., 2020, 25(10): 04020077

Table 6. Summary of *p*-values for model results utilizing LASSO and selection techniques

Variable	Group	LASSO HR	Stepwise HR	Stepwise Sig.
Deck area	Continuous	0.99	0.99	0.03
log(ADTT/lanes on structure)	Continuous	1.04		
Distance from seawater	1 = <0.621 mi (1 km)	1.10		
	2 = >0.621 mi (1 km)	Reference		
Climate zone	1 = Coast	0.90		
	2 = Willamette Valley	Reference		
	3 = Southwestern Valleys	1.08	1.18	0.04
	4 = Northern Cascades			
	5 = High Plateau	0.43	0.00	0.99
	6 = North Central			
	7 = South Central	0.63	0.56	0.00
	8 = Northeast			
	9 = Southeast			
Design period	<1950	1.34	1.60	0.00
	1950–1970	1.07	1.18	0.03
	>1970	Reference		
Maintenance responsibility (NBI Item 21)	State highway	Reference		
	County highway	0.79	0.79	0.02
	City/municipal highway	0.91	0.69	0.11
	State park/forest/ reservation			
	Other state			
	Other local			
	Private			
	Railroad			
Functional classification of inventory route (NBI Item 26)	Principle arterial-interstate (rural)	Reference		
	Principal arterial–other (rural)	0.84	0.68	0.00
	Minor arterial (rural)		0.84	0.10
	Major collector (rural)	0.85	0.66	0.00
	Minor collector (rural)	0.95	0.69	0.01
	Local (rural)	0.88	0.62	0.00
	Principal arterial-interstate (urban)			
	Principal arterial-other freeways (urban)	0.68	0.52	0.03
	Other principal arterial (urban)			
	Minor arterial (urban)			
	Collector (urban)			
	Local (urban)	0.75	0.51	0.08
Kind of material and/or design (NBI Item 43A)	Other	0.49		
	Concrete	Reference		
	Concrete continuous			
	Steel	1.12	1.24	0.02
	Steel continuous			
	Prestressed concrete	0.73	0.72	0.00
	Prestressed concrete continuous	0.99		
	Timber	0.87		
Type of design and/or construction (NBI Item 43B)	Slab	0.86		
	Stringer/multibeam or girder	Reference		
	Girder and floorbeam	0.85	0.63	0.13
	Tee beam			
	Box beam or girders	0.89	0.83	0.09
	Single/spread box beam or girders	0.98		
	frame			
	Deck truss	1.14		
	Thru truss			
	Deck arch			
	Thru arch	1.08		
	Suspension	1.17		
	Lift	0.96		
	Bascule			
	Swing	1.85		
	Culvert			
	Channel beam	1.37	1.88	0.00

Table 6. (*Continued.*)

Variable	Group	LASSO HR	Stepwise HR	Stepwise Sig.
Inspection frequency (NBI Item 91)	6	1.04		
	12	1.35	1.61	0.01
	23			
	24	Reference		
Deck structure type (NBI Item 107)	Cast-in-place	Reference		
	Precast panels	0.82	0.73	0.02
Type of wearing surface (NBI Item 108A)	None	1.30	1.79	0.00
	Monolithic concrete	1.77	2.17	$<2\times10^{-16}$
	Integral concrete	1.51	2.45	0.01
	Latex concrete	2.06	2.37	0.00
	Epoxy overlay	1.87	2.27	0.00
	Bituminous	Reference		
	Gravel			
	Other	0.99		
Type of membrane (NBI Item 108B)	None	Reference		
	Built-up	0.89		
	Preformed fabric	0.90		
	Epoxy	0.98		
	Unknown	0.70		
	Other			
Deck protection (NBI Item 108C)	None	Reference		
	Epoxy	0.97		
	Galvanized			
	Other coated reinforcing			
	Cathodic protection	3.57	4.76	0.00
	Unknown	0.35	0.00	0.98
	Other			

addition, bridge decks inspected every 12 months have an increased risk of deterioration for all models. A shorter inspection period indicates that the maintenance agency is aware of a problem that a bridge deck has and these results thus make sense.

Deck Structure Type (NBI Item 107). Compared with cast-in-place concrete bridge decks, precast concrete bridge decks appear to have a decreased risk of deterioration in all models. This makes sense considering that precast decks are made in a controlled environment with consistent curing resulting in potentially fewer cracking issues.

Type of Wearing Surface (NBI Item 108A). Bridge decks with none, monolithic concrete, integral concrete, latex concrete, and epoxy overlay wearing surfaces have an increased risk of deterioration compared with the ones with bituminous overlay for all models. In addition, bridge decks with other wearing surfaces have a decreased risk of deterioration for the LASSO model. An explanation for these trends could be that inspectors are forced to infer deck condition from the pavement and the soffit as well as other parts of the deck that are visible, unless they employ nondestructive testing tools such as ground penetrating radar (GPR) or impact echo testing (see, e.g., Gucunski et al. 2013), which are still only used in rare cases. Thus, they may not change the CR even if they witness what would be considered normal wear of the asphalt concrete wearing surface (A. Blower, Personal Communication, 2019). In summary, the bridge deck's actual condition is obscured by the presence of a bituminous wearing surface. Therefore, inspectors may be less likely to "downgrade" a deck's CR, which results in the appearance that bridge decks with bituminous wearing surfaces might perform better compared with other wearing surfaces.

Type of Membrane (NBI Item 108B). Bridge decks with built-up, preformed fabric, epoxy, and unknown types of membranes have a decreased risk of deterioration compared with bridge decks with none for the LASSO model. This conclusion makes sense because chloride penetration is supposed to be more difficult when a bridge deck has a membrane.

Deck Protection (NBI Item 108C). The results show that bridge decks with epoxy-coated rebar and unknown deck protection systems have a decreased risk of deterioration compared with bridge decks with none. Bridge decks with cathodic protection have an increased risk of deterioration for all models. These conclusions are consistent with Ghonima et al. (2020), which shows that in general, the presence of deck protection improves bridge deck survival probability. The reason that bridge decks with cathodic protection have an increased risk of deterioration might be that most bridge decks with cathodic protection are located along the coast and may have existing corrosion-related problems.

Survival Curves for Refined Dataset

In order to understand the effect of the design/detailing variables on concrete bridge deck performance, survival curves were created for the independent variables in the Refined Dataset. The survival curves that were created considered all TICR, and not only those with a CR value of 7. This was done so that all the data collected in the Refined Dataset could be captured by the survival curves. Since there is less data in the Refined Dataset than in the NBI Dataset, not all survival curves are significantly different for each variable. Because of this, the survival curves that are not significant should be interpreted as trends. Fig. 3 shows the survival curves for the four design/detailing variables in the Refined Dataset.

While the survival curves for the variable Concrete Cover are not significantly different, bridge decks with a concrete cover <1 in. (25 mm) still have a lower mean survival probability than bridge

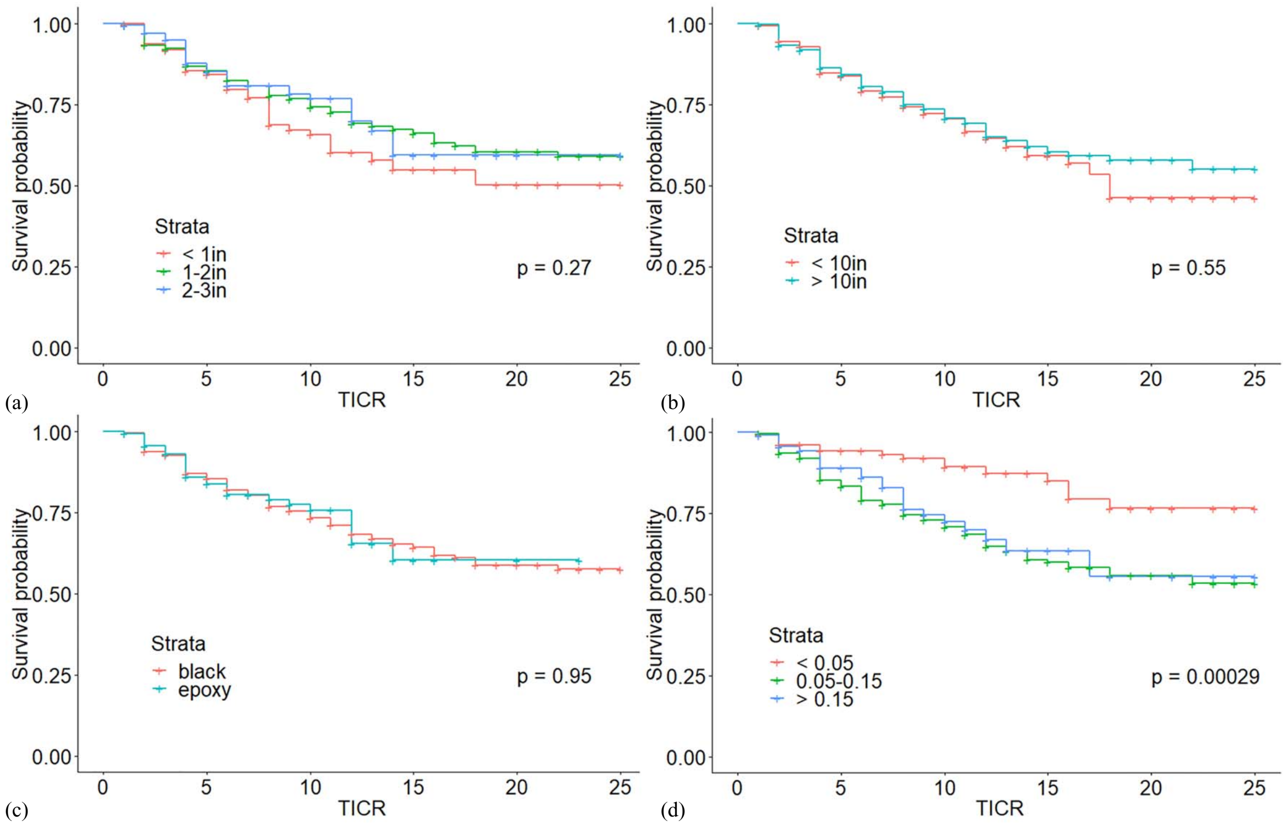

Fig. 3. TICR survival curves depending on select variables for the Refined Dataset: (a) concrete cover; (b) transverse rebar spacing; (c) rebar type; and (d) deck slenderness. Unit conversion: 1 in. = 25 mm.

decks with more cover. This is consistent with the literature, which states that increased cover depth reduces chloride ingress, which in turn decreases bridge deck deterioration (see, e.g., Kirkpatrick et al. 2002; Shi et al. 2012). For the variable Transverse Rebar Spacing, the resulting survival curves show that there is no significant difference between the two curves. However, for TICR > 15 years, bridge decks with a transverse rebar spacing <10 in. (254 mm) exhibit a lower survival probability than bridge decks with rebar spacing >10 in. (254 mm). While a tighter rebar spacing does not prevent cracking, it is still preferred by designers because it leads to more distributed cracking with smaller crack widths. The lower survival probability for TICR > 15 years might be a result of higher stress ranges experienced by these bridges, which leads to increased deterioration in the long term. Although the variable Rebar Type is similar to NBI item 108C Deck Protection, there are some differences between what is reported in the NBI and what was found in the actual bridge drawings from Oregon's BDS. Owing to this inconsistency, the results for the two variables do not agree. The generated survival curves show that there is no significant difference between bridge decks with black rebars and bridge decks with epoxy-coated rebars. The generated survival curves for deck slenderness show that bridge decks with slenderness ≤0.05 have a higher survival probability than other bridge decks. This is plausible considering that bridge decks with a low slenderness are typically precast/prestressed. In addition, the log-rank test shows that at least two of the survival curves are significantly different.

Conclusions

This study utilized a Cox proportional hazards model with LASSO and a backward stepwise-variable selection procedure to study the impact of Oregon NBI variables on concrete highway bridge deck performance. The HR calculated from the NBI Dataset using the LASSO and stepwise models show that bridge decks built before 1970, that are in Climate Zones 2 or 3, and are close to the ocean are all associated with an increased risk of deterioration. In addition, cast-in-place bridge decks on steel bridges without a wearing surface or deck protection have an increased risk of deterioration. In contrast, precast bridge decks in dry climate zones far from the ocean that are on prestressed concrete bridges with deck protection, a membrane, and a bituminous wearing surface have a decreased risk of deterioration. The results from the stepwise model were similar to the results from the LASSO model in that the variables selected for both models had similar HR. However, the LASSO model retained more variables and produced HR with less variation compared with the stepwise model.

In addition, the authors assembled a Refined Dataset comprising design/detailing variables for 400 concrete highway bridge decks in Oregon located in all eight climate zones and distributed across three design periods. Kaplan-Meier survival curves were generated to visualize the impact of design/detailing variables on bridge deck deterioration. For example, bridge decks with a concrete cover <1 in. (25 mm) have a lower survival probability than bridge decks with more cover and bridge decks that are slender have a higher survival probability than other bridge decks.

Moving forward, the impact of the design/detailing variables should be further explored with the availability of more data. Other variables such as in situ construction practice information (e.g., concrete temperature, placement procedure, curing protocol), specifications (e.g., water-to-cement ratio and minimum cementitious material content), and important variables such as deicer use and freeze–thaw cycles could be added to further refine the statistical models and eventually allowing the prediction of the service-life of concrete highway bridge decks. In addition, more refined models should be explored that can capture interaction between independent variables as well as include additional CR.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request. The following items can be requested:

- NBI and Refined Datasets for the State of Oregon

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References

Agrawal, A. K., A. Kawaguchi, and Z. Chen. 2010. "Deterioration rates of typical bridge elements in New York." *J. Bridge Eng.* 15 (4): 419–429. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000123.

Bulusu, S., and K. C. Sinha. 1997. "Comparison of methodologies to predict bridge deterioration." *Transp. Res. Rec.* 1597: 34–42. https://doi.org/10.3141/1597-05.

Fleischhacker, A., O. Ghonima, and T. Schumacher. 2020. "Bayesian Survival Analysis for US Concrete Highway Bridge Decks."*J. Infrastruct. Syst*. 26 (1). https://doi.org/10.1061/(ASCE)IS.1943-555X.0000511.

Friedman, J. H. 2010. Regularization paths for generalized linear models via coordinate descent. *J. Stat. Softw.* 33 (1): 1–22.

FHWA (Federal Highway Administration). 1995. *Recording and coding guide for the structure inventory and appraisal of the Nation's Bridges*. Rep. No. FHWA-PD-96-001. Washington, DC: FHWA.

FHWA (Federal Highway Administration). 2018. *National bridge inventory (NBI)*. Bridges & Structures. Washington, DC: FHWA.

Ghonima, O., J. C. Anderson, T. Schumacher, and A. Unnikrishnan. 2020. "Performance of US concrete highway bridge decks characterized by random parameters binary logistic regression." *J. Risk Uncertainty Eng. Syst. Part A: Civ. Eng.* 6 (1): 04019025. https://doi.org/10.1061/AJRUA6.0001031.

Ghonima, O., T. Schumacher, A. Unnikrishnan, and A. Fleischhacker. 2018. *Advancing bridge technology, task 10: Statistical analysis and modeling of US concrete highway bridge deck performance*. Internal Final Report. FHWA Collaborative Project No. DTFH61-11-H-00027. Washington, DC: FHWA.

Greene, W. H. 1997. *Econometric analysis*. New York: Macmillan.

Gucunski, N., A. Imani, F. Romero, S. Narazian, D. Yuan, H. Wiggenhauser, P. Shokouhi, A. Taffe, and D. Kutrubes. 2013. *Nondestructive testing to identify concrete bridge deck deterioration*. SHRP2 Rep. No. S2-R06A-RR-1. Washington, DC: National Academy of Sciences.

Hatami, A., and G. Morcous. 2011. *Developing deterioration models for Nebraska Bridges*. Lincoln, NE: Nebraska Dept. of Roads.

Kaplan, E. L., and P. Meier. 1958. "Nonparametric estimation from incomplete observations." *J. Am. Stat. Assoc.* 53 (282): 457–481.

Kim, Y. J., and D. K. Yoon. 2010. "Identifying critical sources of bridge deterioration in cold regions through the Constructed Bridges in North Dakota." *J. Bridge Eng.* 15 (5): 542–552. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000087.

Kirkpatrick, T. J., R. E. Weyers, C. M. Anderson-Cook, and M. M. Sprinkel. 2002. "Probabilistic model for the chloride-induced corrosion service life of bridge decks." *Cem. Concr. Res.* 32 (12): 1943–1960. https://doi.org/10.1016/S0008-8846(02)00905-5.

Leung, K.-M., R. M. Elashoff, and A. A. Afifi. 1997. "Censoring issues in survival analysis." *Ann. Rev. Public Health* 18 (1): 83–104. https://doi.org/10.1146/annurev.publhealth.18.1.83.

Madanat, S., R. Mishalani, and W. H. W. Ibrahim. 1995. "Estimation of infrastructure transition probabilities from condition rating data." *J. Infrastruct. Syst.* 1 (2): 120–125. https://doi.org/10.1061/(ASCE)1076-0342(1995)1:2(120).

Madanat, S. M., M. G. Karlaftis, and P. S. McCarthy. 1997. "Probabilistic infrastructure deterioration models with panel data." *J. Infrastruct. Syst.* 3 (1): 4–9. https://doi.org/10.1061/(ASCE)1076-0342(1997)3:1(4).

Mauch, M., and S. Madanat. 2001. "Semiparametric hazard rate models of reinforced concrete bridge deck deterioration." *J. Infrastruct. Syst.* 7 (2): 49–57. https://doi.org/10.1061/(ASCE)1076-0342(2001)7:2(49).

Mishalani, R. G., and S. M. Madanat. 2002. "Computation of infrastructure transition probabilities using stochastic duration models." *J. Infrastruct. Syst.* 8 (4): 139–148. https://doi.org/10.1061/(ASCE)1076-0342(2002)8:4(139).

Morcous, G. 2006. "Performance prediction of bridge deck systems using Markov Chains." *J. Perform. Constr. Facil* 20 (2): 146–155. https://doi.org/10.1061/(ASCE)0887-3828(2006)20:2(146).

Morcous, G., Z. Lounis, and M. S. Mirza. 2003. "Identification of enviromental categories for Markovian deterioration models of bridge decks." *J. Bridge Eng.* 8 (6): 353–361. https://doi.org/10.1061/(ASCE)1084-0702(2003)8:6(353).

Nasrollahi, M., and G. Washer. 2015. "Estimating inspection intervals for bridges based on statistical analysis of national bridge inventory data." *J. Bridge Eng.* 20 (9): 04014104. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000710.

Rich, J. T., J. G. Neely, R. C. Paniello, C. C. Voelker, B. Nussenbaum, and E. W. Wang. 2010. "A practical guide to understanding Kaplan-Meier curves." *Otolaryngol Head Neck Surg.* 143 (3): 331–336.

Setunge, S., and M. S. Hasan. 2011. *Concrete bridge deterioration prediction using Markov Chain approach*. Melbourne, Australia: RMIT Univ.

Shi, X., N. Xie, K. Fortune, and J. Gong. 2012. "Durability of steel reinforced concrete in chloride environments: An overview." *Constr. Build. Mater.* 30: 125–138. https://doi.org/10.1016/j.conbuildmat.2011.12.038.

Sobanjo, J., P. Mtenga, and M. Rambo-Roddenberry. 2010. "Reliability-based modeling of bridge deterioration hazards." *J. Bridge Eng.* 15 (6): 671–683. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000074.

Stewart, M. G., and D. V. Rosowsky. 1998. "Structural safety and serviceability of concrete bridges subject to corrosion." *J. Infrastruct. Syst.* 4 (4): 146–155. https://doi.org/10.1061/(ASCE)1076-0342(1998)4:4(146).

Sullivan, L. 2016. "Survival analysis." Boston Univ. School of Public Health. Accessed June 3, 2016. http://sphweb.bumc.bu.edu/otlt/MPH-Modules/BS/BS704_Survival/BS704_Survival_print.html.

Tabatabai, H., C.-W. Lee, and M. A. Tabatabai. 2016. "Survival analyses for bridge decks in Northern United States." *Civ. Environ. Eng.* 7: 1–30.

Tabatabai, H., M. Tabatabai, and C.-W. Lee. 2011. "Reliability of bridge decks in Wisconsin." *J. Bridge Eng.* 16 (1): 53–62. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000133.

Taylor, G. H., and C. Hannan. 1999. *The climate of oregon: From rain forest to desert*. Corvallis, OR: Oregon State University Press.

Tibshirani, R. 1996. Regression shrinkage and selection via the LASSO. *J R Stat Soc Series B Stat Methodol.* 58 (1): 267–288. www.jstor.org/stable/2346178.

Tibshirani, R. 1997. The LASSO method for variable selection in the Cox model. *Stat. Med.* 16: 385–395.

Venables, W. N., and B. D. Ripley. 2002. *Modern applied statistics with S*. New York: Springer.

Vu, K. A., and M. G. Stewart. 2000. "Structural reliability of concrete bridges including improved chloride-induced corrosion models." *Struct. Saf.* 22 (4): 313–333. https://doi.org/10.1016/S0167-4730(00)00018-7.